PROJECT REPORT - CUSTOM GREP

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1. Introduction

This project involves developing simplified UNIX-like utilities in C along with a custom shell. In this part, I implemented custom_grep, which searches for a pattern in a file and prints matching lines.

2. Objective

- Implement grep-like functionality in C
- Understand file reading using fgets()
- Perform string matching using strstr()
- Handle command-line arguments and flags (-i)
- Implement basic error handling

3. Features of custom_grep

- Search for a pattern in a file
- Print matching lines with line numbers
- Case-insensitive search using -i flag
- Simple and efficient implementation
- Error handling for missing or inaccessible files

4. Concepts Used

- File I/O (fopen, fgets, fclose)
- String handling (strstr, strcpy)
- Character handling (tolower)
- Command-line arguments (argc, argv)
- Error handling using perror()

5. Code Explanation

The program reads the file line by line using fgets(). It checks whether each line contains the given pattern using strstr(). For case-insensitive search, both the line and pattern are converted to lowercase before matching. If a match is found, the line along with its line number is printed.

6. Execution Steps

Compile:
gcc -Wall -g -o custom_grep custom_grep.c

Run standalone:
./custom_grep pattern file.txt

Run inside custom shell:
./custom_shell
custom_grep pattern file.txt

7. Usage Examples

./custom_grep hello file.txt
./custom_grep -i hello file.txt

8. Screenshots

Test 1 — Basic search: custom_grep hello file.txt

```
custom_shell:/mnt/d/os project 2/G25_OS-project/G25_Project2_1$ custom_grep hello file.txt
3: hello this is os project
custom_shell:/mnt/d/os project 2/G25_OS-project/G25_Project2_1$
```

Test 2 — Case-insensitive search: custom_grep -i hello file.txt

```
custom_shell:/mnt/d/os project 2/G25_OS-project/G25_Project2_1$ custom_grep -i hello file.txt
1: Hello world
3: hello this is os project
custom_shell:/mnt/d/os project 2/G25_OS-project/G25_Project2_1$
```

Test 3 — File not found: custom_grep hello file1.txt

```
custom_shell:/mnt/d/os project 2/G25_OS-project/G25_Project2_1$ custom_grep hello file1.txt
Error opening file: No such file or directory
custom_shell:/mnt/d/os project 2/G25_OS-project/G25_Project2_1$
```

9. Conclusion

This project helped in understanding file handling, string operations, and command-line argument parsing in C. It also demonstrated how basic UNIX utilities work internally.